

# Reservoir Architecture

## Baseline temporal memory

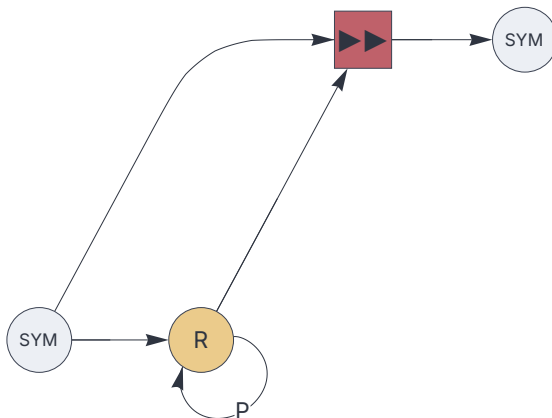
A temporal memory, consisting of a simple recurrent delay node and a hetero-associative predictor node. This model learns sequences and next-token prediction on the fly.

## Configuration

```
<< "Circuits.m"
```

```
dataflow = {
  input[codec] [11] [],
  delay["R"→60] [51] [{11,51"P"}],
  predictor[] [101] [11, 51],
  output[codec] [] [101],
  _ → {2000, 10}
};
```

```
c = Circuit[dataflow]; c[Graph, ImageSize→ 300]
```



## Synthetic training data

Generate a 2000 character random string.

```
chars = CharacterRange["A", "Z"]
```

```
{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z}
```

```
dataset = RandomChoice[ chars, 2000];
```

## Analyze prediction score

```
Do[
  predictions = c /@ dataset;
  Print["    iteration = ", i,
    ", prediction score = ", N[Mean@Boole@MapThread[SameQ,
      {dataset, RotateRight[predictions]}]]],
  {i, 1, 10}];
```

```
iteration = 1, prediction score = 0.0255
```

```
iteration = 2, prediction score = 0.8475
```

```
iteration = 3, prediction score = 0.981
```

```
iteration = 4, prediction score = 0.996
```

```
iteration = 5, prediction score = 0.9985
```

```
iteration = 6, prediction score = 0.9985
```

```
iteration = 7, prediction score = 0.998
```

```
iteration = 8, prediction score = 0.999
```

```
iteration = 9, prediction score = 1.
```

```
iteration = 10, prediction score = 1.
```